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1. 习题七:

5.

$$L(\lambda) = \prod_{i=1}^{10} \lambda e^{-\lambda x_i}$$

$$\ln L(\lambda) = \sum_{i=1}^{10} (\ln \lambda - \lambda x_i) = 10 \ln \lambda - \lambda \sum_{i=1}^{10} x_i$$

$$\text{故令 } \frac{\partial L(\lambda)}{\partial \lambda} = \frac{10}{\lambda} - \sum_{i=1}^{10} x_i = 0$$

$$\lambda = \frac{10}{\sum_{i=1}^{10} x_i} = 0.0008562$$

故 λ 的最大似然估计值为 0.0008562

6.

(1) 由题意得

$$E(X) = kp = \frac{1}{n} \sum_{i=1}^n X_i$$

解得 p 的矩估计量为 $\hat{p} = \frac{\bar{X}}{k}$

总体 X 的分布律为 $P(X = x; p) = C_k^x p^x (1-p)^{k-x}, x = 0, 1, \dots, k$

设 $x_1, \dots, x_n$ 为样本的一组观测值, 则

$$\text{似然函数为 } L(p) = \prod_{i=1}^n C_k^{x_i} p^{x_i} (1-p)^{k-x_i}$$

$$\ln L(p) = \sum_{i=1}^n C_k^{x_i} + x_i \ln p + (k - x_i) \ln(1-p)$$

$$\text{令 } \frac{d \ln L(p)}{dp} = \sum_{i=1}^n \frac{x_i}{p} - \frac{k - x_i}{1-p} = \frac{1}{p} \sum_{i=1}^n x_i - \frac{nk - \sum_{i=1}^n x_i}{1-p} = 0$$

解得 p 的最大似然估计值为 $\hat{p} = \frac{\bar{x}}{k}$, 最大似然估计量为 $\hat{p} = \frac{\bar{X}}{k}$

$$(3) E(X) = \int_{-\infty}^{+\infty} x f(x; \theta) dx = \int_0^1 \sqrt{\theta} x^{\sqrt{\theta}} dx = \frac{\sqrt{\theta}}{\sqrt{\theta}+1} = \bar{X}$$

$$\text{故解得 } \hat{\theta} = \frac{\bar{X}^2}{(1-\bar{X})^2}$$

设 $x_1, \dots, x_n$ 为样本的一组观测值, 则似然函数为 $L(\theta) = \prod_{i=1}^n \sqrt{\theta} x_i^{\sqrt{\theta}-1}$

$$\text{故 } \ln L(\theta) = \sum_{i=1}^n \frac{1}{2} \ln \theta + (\sqrt{\theta} - 1) \ln x_i$$

$$\text{令 } \frac{d \ln L(\theta)}{d\theta} = \sum_{i=1}^n \frac{1}{2\theta} + \frac{\ln x_i}{2\sqrt{\theta}} = \frac{n}{2\theta} + \frac{\sum_{i=1}^n \ln x_i}{2\sqrt{\theta}} = 0$$

解得 θ 的最大似然估计值为 $\hat{\theta} = \left(\frac{n}{\sum_{i=1}^n \ln x_i} \right)^2$

2 故 θ 的最大似然估计量为 $\hat{\theta} = (\frac{n}{\sum_{i=1}^n \ln X_i})^2$

(5) 由题意得

$$\begin{cases} E(X) = \int_{-\infty}^{+\infty} xf(x; \theta)dx = \int_{\mu}^{+\infty} \frac{x}{\theta} e^{-\frac{x-\mu}{\theta}} dx = \theta + \mu = \bar{X} \\ E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x; \theta)dx = \int_{\mu}^{+\infty} \frac{x^2}{\theta} e^{-\frac{x-\mu}{\theta}} dx = \mu^2 + 2\theta\mu + 2\theta^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 \end{cases} \quad (1)$$

解得 θ, μ 的矩估计量为 $\hat{\theta} = \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2 - (\frac{1}{n} \sum_{i=1}^n X_i)^2}$, $\hat{\mu} = \bar{X} - \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2 - (\frac{1}{n} \sum_{i=1}^n X_i)^2}$

设 $x_1, \dots, x_n$ 为样本的一组观测值, 则似然函数为 $L(\theta, \mu) = \prod_{i=1}^n \frac{1}{\theta} e^{-\frac{x_i - \mu}{\theta}}$

故 $\ln L(\theta, \mu) = \sum_{i=1}^n (-\ln \theta - \frac{x_i - \mu}{\theta})$

$$\begin{cases} \frac{\partial \ln L(\theta, \mu)}{\partial \theta} = \sum_{i=1}^n (-\frac{1}{\theta} + \frac{x_i - \mu}{\theta^2}) = \frac{\sum_{i=1}^n x_i - n\mu}{\theta^2} - \frac{n}{\theta} = 0 \\ \frac{\partial \ln L(\theta, \mu)}{\partial \mu} = \sum_{i=1}^n \frac{1}{\theta} > 0 \end{cases} \quad (2)$$

故 $L(\theta, \mu)$ 对于 μ 单调递增, 要使 (θ, μ) 尽可能大, 则 μ 尽可能大. 又因为 $\mu \leq x_1, \dots, \mu \leq x_n$

故取 μ 的最大似然估计量为 $\hat{\mu} = \min\{X_1, \dots, X_n\} = X_{(1)}$

又解得 θ 的最大似然估计值为 $\hat{\theta} = \bar{x} - \mu$

故 θ 的最大似然估计量为 $\hat{\theta} = \bar{X} - X_{(1)}$

7.

令 $Y = \ln X, Y \sim N(\mu, \sigma), X = e^Y$

故 Y 的密度函数为 $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$

$E(X) = E(e^Y) = \int_{-\infty}^{+\infty} e^y \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy = e^{\mu + \frac{\sigma^2}{2}}$

$E(X^2) = E(e^{2Y}) = \int_{-\infty}^{+\infty} e^{2y} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy = e^{2\mu + 2\sigma^2}$

故 $D(X) = E(X^2) - E(X)^2 = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}$

又因为 $Y \sim N(\mu, \sigma^2)$, 故 μ, σ^2 的最大似然估计量为

$\hat{\mu} = \bar{Y} = \frac{1}{n} \sum_{i=1}^n X_i, \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2 = \frac{1}{n} \sum_{i=1}^n (\ln X_i - \frac{1}{n} \sum_{i=1}^n \ln X_i)^2$

由最大似然估计的不变性原理, $E(X), D(X)$ 的最大似然估计量为

$\hat{E}(X) = e^{\hat{\mu} + \frac{\hat{\sigma}^2}{2}}, \hat{D}(X) = e^{2\hat{\mu} + 2\hat{\sigma}^2} - e^{2\hat{\mu} + \hat{\sigma}^2}$

2. 补充题:

$$(1) L(\mu_1, \mu_2, \sigma^2) = \prod_{i=1}^m \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu_1)^2}{2\sigma^2}} \times \prod_{j=1}^n \frac{1}{2\sqrt{\pi}\sigma} e^{-\frac{(y_j - \mu_2)^2}{8\sigma^2}}$$

$$(2) \text{ 设 } x_1, \dots, x_m, y_1, \dots, y_n \text{ 为样本的一组观测值, 则 } \ln L(\mu_1, \mu_2, \sigma^2) = \sum_{i=1}^m (-\ln\sqrt{2\pi} - \frac{1}{2}\ln\sigma^2 - \frac{(x_i - \mu_1)^2}{2\sigma^2}) + \sum_{j=1}^n (-\ln 2\sqrt{\pi} - \frac{1}{2}\ln\sigma^2 - \frac{(y_j - \mu_2)^2}{8\sigma^2})$$

$$\begin{cases} \frac{\partial \ln L(\mu_1, \mu_2, \sigma^2)}{\partial \mu_1} = \sum_{i=1}^m \frac{x_i - \mu_1}{\sigma^2} = 0 \\ \frac{\partial \ln L(\mu_1, \mu_2, \sigma^2)}{\partial \mu_2} = \sum_{j=1}^n \frac{y_j - \mu_2}{4\sigma^2} = 0 \\ \frac{\partial \ln L(\mu_1, \mu_2, \sigma^2)}{\partial \sigma^2} = \sum_{i=1}^m \left(-\frac{1}{2\sigma^2} + \frac{(x_i - \mu_1)^2}{2\sigma^4}\right) + \sum_{j=1}^n \left(-\frac{1}{2\sigma^2} + \frac{(y_j - \mu_2)^2}{8\sigma^4}\right) = 0 \end{cases}$$

$$\text{解得 } \hat{\mu}_1 = \bar{X}, \hat{\mu}_2 = \bar{Y}, \hat{\sigma}^2 = \frac{\sum_{i=1}^m (X_i - \bar{X})^2 + \sum_{j=1}^n (Y_j - \bar{Y})^2}{m+n}$$